# CPE 334 Introduction to Software Engineering

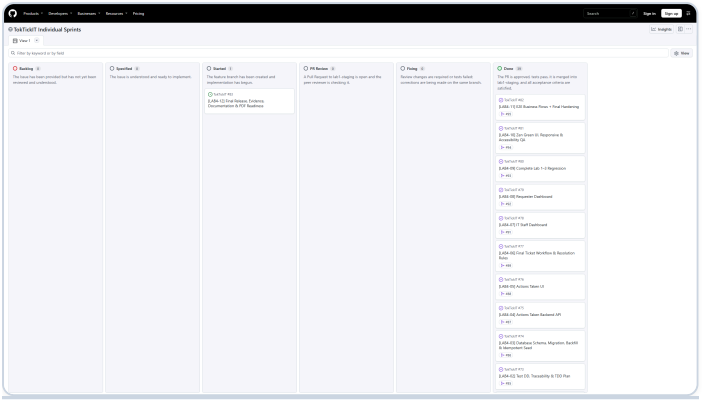
Lab 4: TokTickIT Actions Taken, Dashboards, and Final Regression (Semester 1/2026)

**Author:** Aphichaya Klinhual (67070503447) • @Apichaya251400
**Peer Reviewer:** Pilaiwan Churdchu (67070503492) • @Pilaiwan3492
**Repository:** github.com/Apichaya251400/TokTickIT

## Answer Part 1: Git Use with Engineering Workflow

10 Points

**Engineering Process & Non-Self-Merge Rule:** Every sprint feature was planned as an independent GitHub Issue (Issues #72–#83) on the Kanban Project Board, developed on an isolated `feature/*` branch, verified with local test suites, pushed to GitHub, and submitted as a formal Pull Request to `lab4-staging`. Under the strict course governance constraint ("*ห้าม merge เองเด็ดขาด*"), **zero PRs were self-merged**. Every PR underwent mandatory peer review by collaborator **Pilaiwan Churdchu (@Pilaiwan3492)**, who reviewed code, tested edge cases, requested changes where necessary, approved, and merged each PR.



EVD-REL-01: GitHub Project Kanban Board (Project #1) with all completed Lab 4 Issues (#72..#82) in Done column.




EVD-REL-02: Git Commit Tree & Release Branching Graph showing clean feature branch merges into `lab4-staging`.

# Lab 4 — Peer Review Record (EVD-REL-03)

**Repository**: https://github.com/Apichaya251400/TokTickIT | **Branch**: `feature/lab4-release`

**Reviewer**: Pilaiwan Churdchu (67070503492 / @Pilaiwan3492) | **Author**: Aphichaya Klinhual (67070503447 / @Apichaya251400)

## Reviews Received (Apichaya251400/TokTickIT)

| PR | Review Outcome | Review Comment Received | Author Response |
|---|---|---|---|
| **#84** (LAB4-01) | Approved | Clarify definitions of operational dashboards, timezone UTC+7, createdAt authoritative. | Clarified in spec, merged into lab4-staging. |
| **#85** (LAB4-02) | Approved | Verified test strategy against requirements. 0 blocking issues. | Merged into lab4-staging. |
| **#86** (LAB4-03) | Approved | Migration replay caught silently ignoring errors; restrict to duplicate objects. | Updated migration.test.ts, merged. |
| **#87** (LAB4-04) | Approved | Align assigneeId to integer, add inactive-assignee rejection test. | Aligned and added TEST-ACT-06, merged. |
| **#88** (LAB4-05) | Approved | Improve keyboard trap, retry state on error, verify ticket sequence sync. | Added focus trap, retry state, sequence sync. |
| **#89** (LAB4-06) | Approved | Verified 15 transitions, 409 Conflict handling, Resolution Gate validation. | Merged into lab4-staging. |
| **#91** (LAB4-07) | Approved | Document GET /api/tickets authorization rules (Requester 403, Admin allowed). | Documented in api-spec.md, merged. |
| **#92** (LAB4-08) | Approved | Align 30-day recently resolved filter contract, fix not-found navigation. | Aligned filter, added EVD-REQBDASH-08, merged. |
| **#93** (LAB4-09) | Approved | Playwright in CI, 4-card responsive check, visible badges, product hardening suite. | Implemented 483 tests, split clean logs, hardening suite. |
| **#94** (LAB4-10) | Approved | Rendered UI spec, dynamic assertions, 44x44px touch targets, 2px focus ring. | Addressed 7 points, 100% green CI. |
| **#95** (LAB4-11) | Approved | ACT-07 verify ticket state, dynamic snapshot & restore updatedAt via executeRaw. | Resolved all 5 items, merged commit 1cdc4a6. |
| **#96** (LAB4-12) | In Review | Final release gate, cross-document audit, Part 1–9 submission preparation. | Submitted on feature/lab4-release, awaiting peer review. |

## Reviews Given (Pilaiwan3492/TokTickIT)

| PR Reviewed | Comment Given | Partner Response |
|---|---|---|
| **#82** (PR 30) | Clarify IT Staff current-user actions on dashboard, define recently resolved tickets. | Updated specification, merged into lab4-staging. |
| **#83** (PR 31) | Expand regression to Labs 1–3, add mobile screenshot inventory, fix test paths. | Expanded regression scenarios, merged. |
| **#84** (PR 32) | Approved! Database schema, migration, and idempotent seed match requirements. | Merged into lab4-staging. |
| **#85** (PR 33) | Approved! Actions Taken backend REST APIs, idempotency, and concurrency pass. | Merged into lab4-staging. |
| **#86** (PR 34) | Align Resolution Gate in BR-11 with BR-09, tighten ISO 8601 offset regex. | Aligned BR-11, tightened regex, added boundary tests. |
| **#87** (PR 35) | Verify conflict-recovery flow in ActionTakenModal with latest updatedAt. | Updated conflict-recovery flow, added error tests. |
| **#88** (PR 36) | Verify priority=URGENT drilldown matches metric, verify queue filters reset. | Synchronized parameter, reset filters on query removal. |
| **#89** (PR 37) | Update tests.md statuses, verify drilldown destination filters and empty states. | Updated tests.md, added drilldown assertions. |
| **#90** (PR 38) | Align Ticket 8 seed fixture with E2E reset helper, separate Labs 1–3 regression. | Aligned seed comment with davidStaffId owner. |
| **#91** (PR 39) | Reviewed multi-device regression suite, client tests for ticket lifecycle & submission. | Partner marked in-progress; awaiting final update. |

EVD-REL-03: Rendered `docs/lab-04/reviewer.md` showing Peer Review dialogue, reviewer identity, PR links, and approval states.




EVD-REL-06: Clean Repository Directory Structure showing `server/`, `client/`, `e2e/`, `docs/lab-04/`, and `artifacts/lab-04/`.

### Repository Integrity & .gitignore Audit

- **README.md**: Fully maintained in repo root with prerequisites (Node 20+, Postgres 16), setup guide, seed credentials table, dev server commands, and 485-test summary table.
- **.gitignore**: Clean rules ignoring `node_modules/`, build artifacts `dist/`, local env files `.env`, and database binaries. Explicitly whitelists `!artifacts/**/*.log` ensuring zero evidence suppression.
- **Peer Review Evidence**: Total of 12 PRs reviewed across author and reviewer repositories (Author PRs | Reviewer PRs).

## Answer Part 2: Spec DD

5 Points

**Specification-Driven Development (SpecDD):** The engineering contract was formally specified, reviewed, and merged in PR #84 (Issue #72 / LAB4-01) *prior to implementing code PRs*. All downstream models, APIs, UI workflows, and test assertions strictly conform to these numbered contracts:

- **15 Functional Requirements (FR-01..15)**: Operational dashboards, Actions Taken lifecycle, Resolution Gate, audit logging, and role-based permissions (docs/lab-04/specification.md).
- **17 Business Rules (BR-01..17)**: State machine rules (15 allowed transitions, 4 terminal blocks), resolution eligibility (>= 1 completed action or mandatory resolution note), optimistic concurrency ( `409 STALE_UPDATE` ), and deduplication ( `Idempotency-Key` ).
- **23 Acceptance Criteria (AC-01..23)**: Authoritative operational metric calculations, role isolation, multi-viewport responsive guarantees, and visual accessibility standards.
- **Database & Migration Decisions**: Added `ActionTaken` and `IdempotencyRecord` models in PostgreSQL; pure idempotent SQL migrations; dynamic synchronization of `ticket_number_seq` in seed.




EVD-REL-04: Sprint 4 Engineering Contract Package Audit ( `specification.md` , `api-spec.md` , `ui-spec.md` , `tests.md` ) showing 100% bidirectional traceability.

## Answer Part 3: Test DD and Traceability

10 Points

**Test-Driven Development & Traceability:** Every requirement, business rule, and acceptance criterion is mapped to automated test IDs in docs/lab-04/tests.md. All 21 referenced test files physically exist on disk, achieving **100% bidirectional coverage**:

| Test Suite / Layer | Tooling & Environment | Files | Tests Run | Passed | Status |
|---|---|---|---|---|---|
| **Server API & Integration** | Vitest + Supertest + PostgreSQL | 29 | 284 | **284** | 100% PASS |
| **Client Component UI** | Vitest + React Testing Library + jsdom | 17 | 129 | **129** | 100% PASS |
| **Playwright E2E Hardening** | Playwright Chromium (Desktop, Tablet, Mobile) | 13 | 72 | **72** | 100% PASS |
| **Production Build** | TypeScript `tsc` + Vite production bundle | 2 | 2 | **0 errors** | 100% PASS |
| **Integrated Total** | Unified Full-Stack Test Suite | **59** | **485** | **485 / 485** | 100% GREEN |

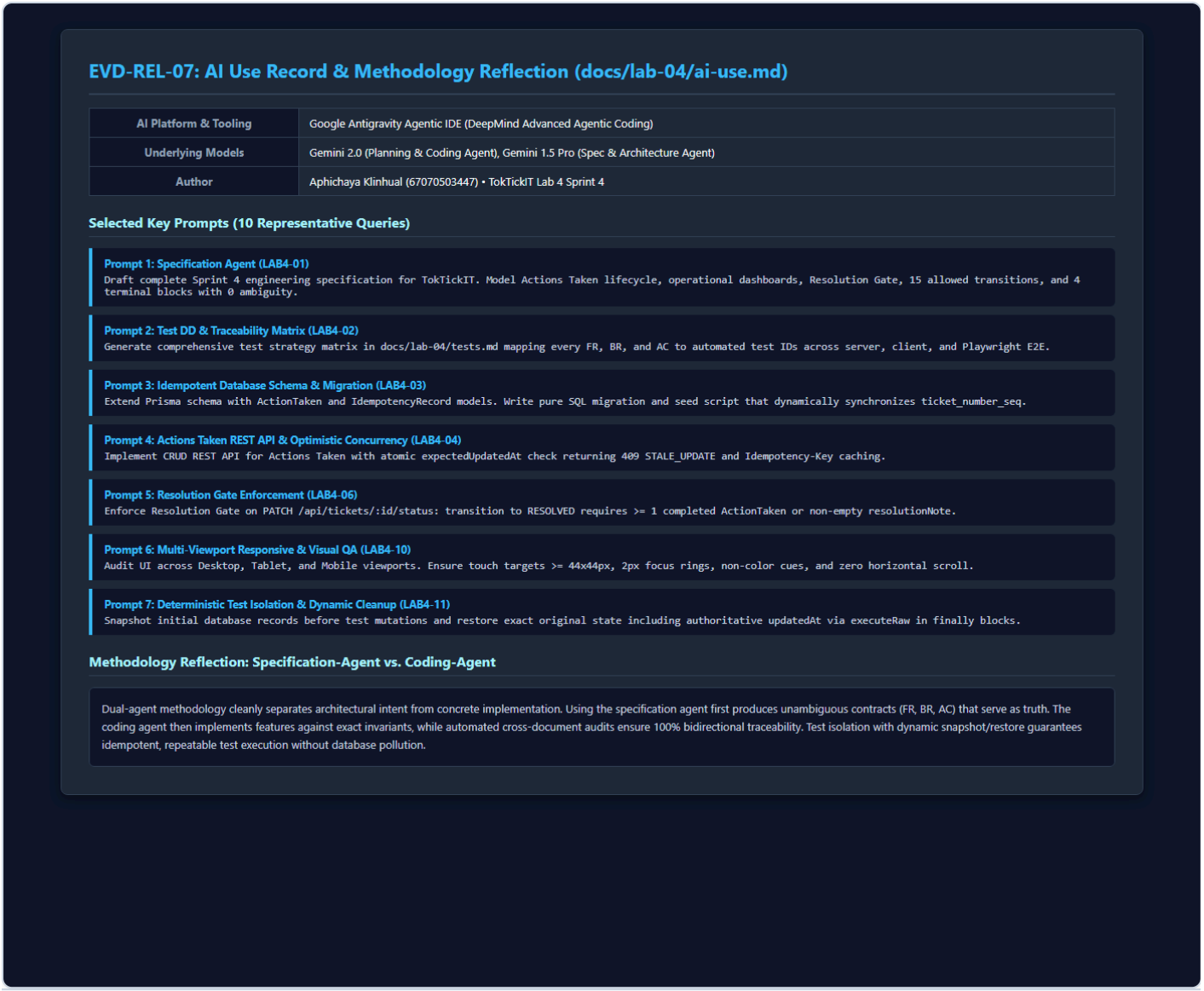
**Traceability & Deterministic Test Isolation Summary (EVD-REL-08 & EVD-REL-09)**

• **Zero Undefined IDs**: 15/15 FRs, 17/17 BRs, 23/23 ACs mapped with 0 missing requirements and 0 broken relative links.

• **Authoritative Database Restoration**: Mutating E2E tests ( `TEST-CONC-01` , `E2E-FLOW-07` , `E2E-FLOW-08` ) dynamically capture pre-test record state and restore exact original values including authoritative `updatedAt` via raw SQL in `finally` blocks, guaranteeing repeatable, idempotent test runs.

## Answer Part 4: AI Use with Reflection

5 Points




EVD-REL-07: AI Use Record & Methodology Reflection (docs/lab-04/ai-use.md).

### My Reflection: Specification-Agent vs. Coding-Agent Workflow

In Lab 4, utilizing **Google Antigravity Agentic IDE** powered by **Gemini 2.0** (Coding & Planning Agent) and **Gemini 1.5 Pro** (Specification & Architectural Agent) reinforced the necessity of strict workflow separation between specification and execution:

1. **Specification-First Prevents Rework**: Treating the Specification Agent as a formal systems architect produced unambiguous contracts (FR-01..15, BR-01..17, AC-01..23). In Sprint 4, explicitly defining the 15 valid status transitions and 4 terminal blocks upfront eliminated race conditions and conflicting UI states during implementation.

2. **Coding Agent as Rigorous Pair Programmer**: The Coding Agent implemented atomic Prisma database transactions, optimistic concurrency protection ( `409 STALE_UPDATE` ), and header-based idempotency. Peer review feedback from collaborator Pilaiwan Churdchu identified subtle issues (e.g. database `updatedAt` mutation during cleanup and missing Ticket state assertions in ACT-07), which the Coding Agent resolved by introducing dynamic snapshotting and raw SQL restoration in test cleanup.

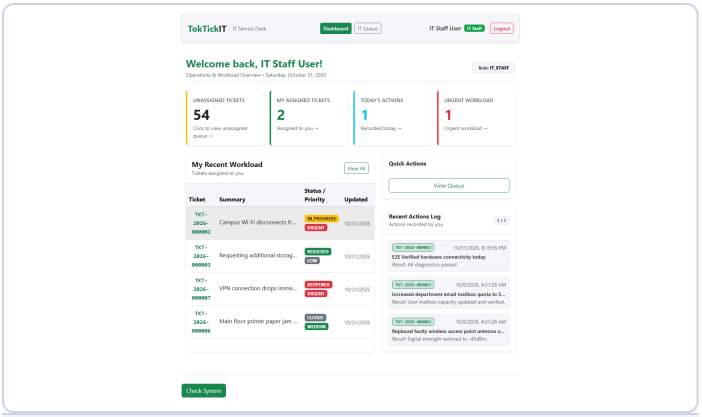
3. **Traceability & Continuous Auditing**: Automated cross-document audit scripts ensured zero drift between `specification.md` , `api-spec.md` , `ui-spec.md` , and `tests.md` , proving that AI assistance is most potent when bound to machine-verifiable engineering gates.

## Answer Part 5: Working IT Staff Dashboard UI

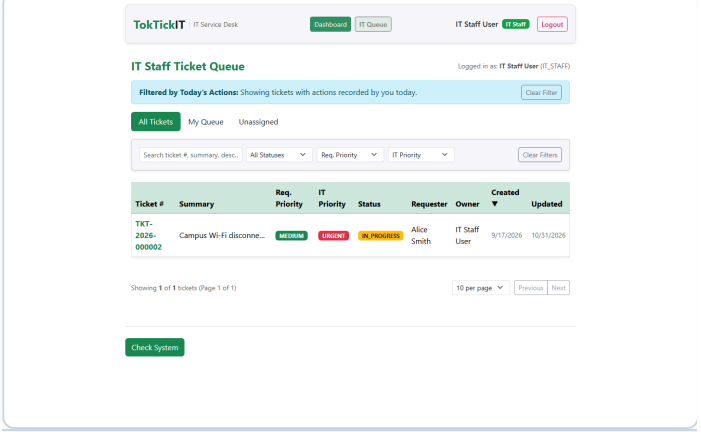
5 Points

**IT Staff Dashboard (StaffDashboard.tsx / `GET /api/dashboards/staff` ):** Displays 4 authoritative operational metric cards (Unassigned Active, My Active Workload, Actions Recorded Today, Urgent Unresolved), quick action shortcuts, recent tickets,

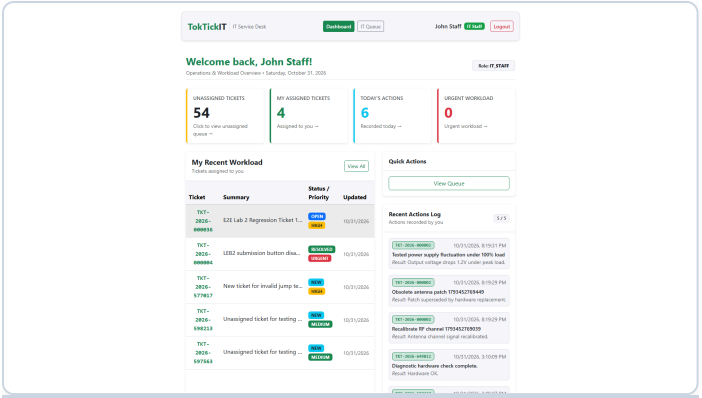
and drill-down filters to the queue:



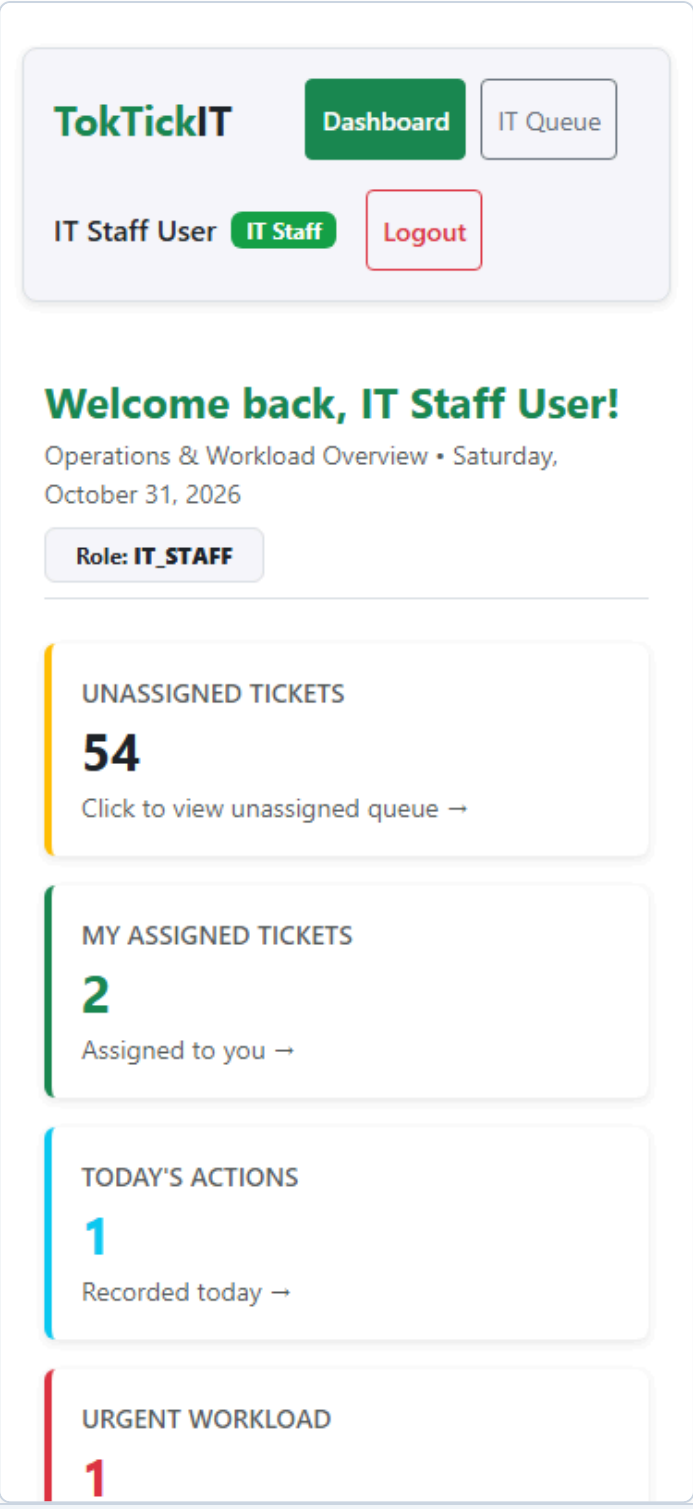

EVD-STAFFDASH-01: Operational IT Staff Dashboard on Desktop (1920×1080) with 4-card metrics grid and recent ticket worklist.




EVD-STAFFDASH-05: Queue Drill-Down Navigation filtering by metric criteria (e.g. Unassigned, My Workload, Urgent).




EVD-E2E-04: Authoritative Backend Calculation Verification matching database queries with 0 discrepancy.

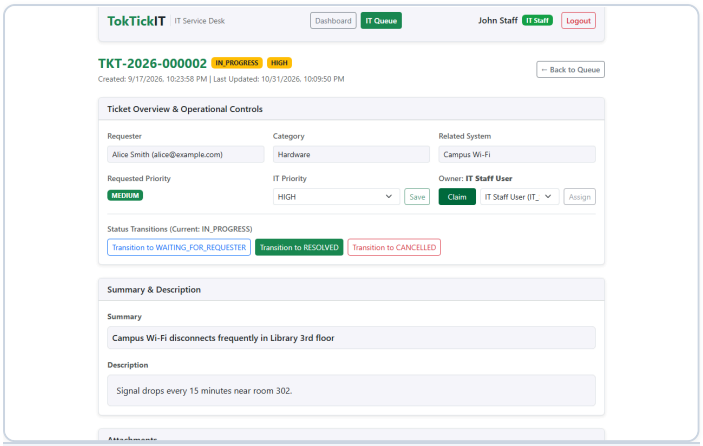



EVD-STAFFDASH-02-mobile: Mobile Layout (375×812) adapting into single-column vertical stack with touch targets >= 44×44px.

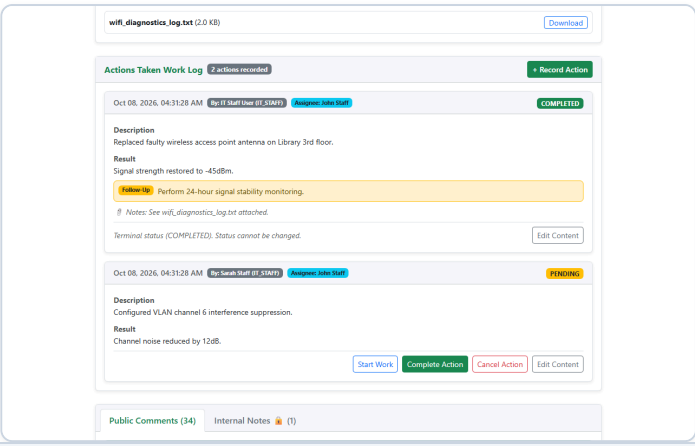
## Answer Part 6: Working Actions Taken UI

**10 Points**

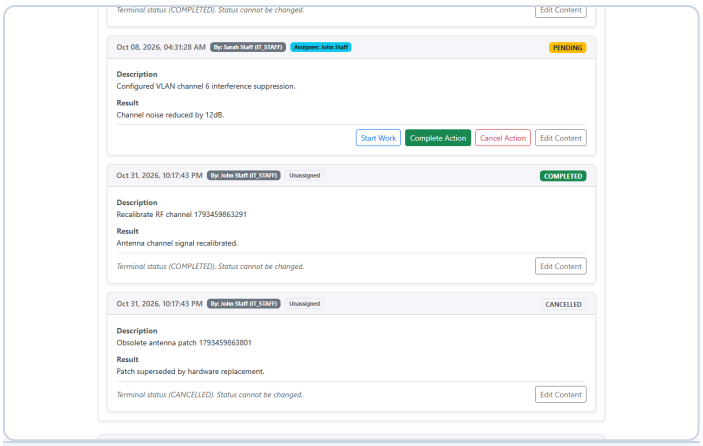
**Actions Taken Workspace (ActionsTakenWorkspace.tsx / `/api/tickets/:id/actions`):** IT Staff record technical diagnosis, assignees, duration, status progression ( `PENDING` $ o$ `IN_PROGRESS` $ o$ `COMPLETED` / `CANCELLED` ), attachments, and follow-up notes. Requesters have strict read-only access (AC-04):



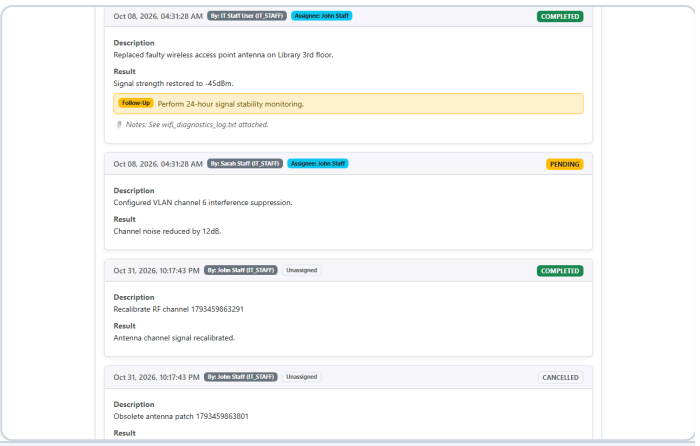

EVD-ACT-01: Desktop Actions Taken List showing action cards, assigned staff, status badges, and duration.



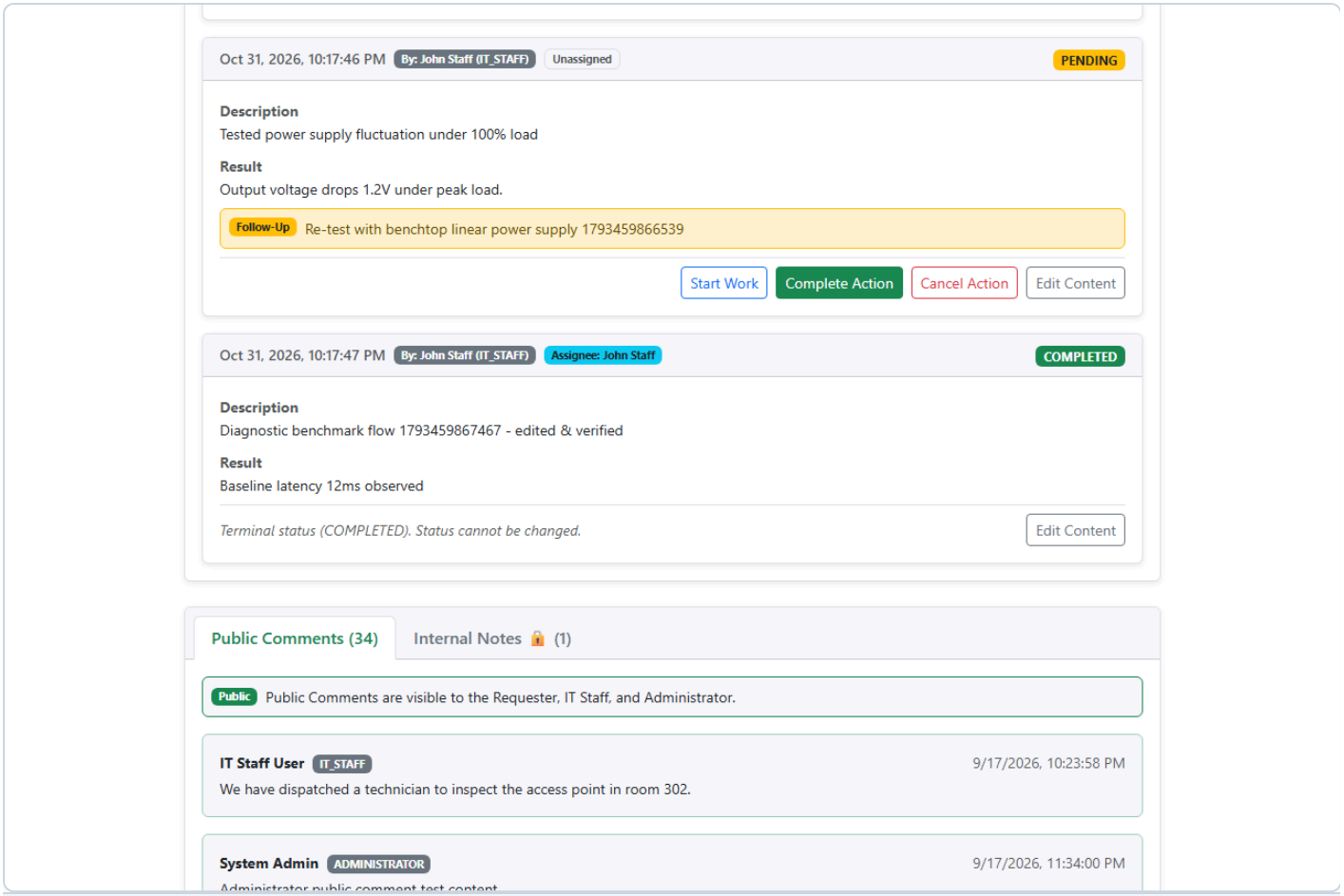

EVD-ACT-03: Multiple Actions Taken on one Ticket demonstrating stable chronological ordering and audit history.




EVD-ACT-04: Action Edit Modal with status transitions (Complete/Cancel), validation rules, and follow-up notes.




EVD-ACT-05: Inactive Assignee Rejection alert and Requester read-only view (Create Action button strictly hidden).

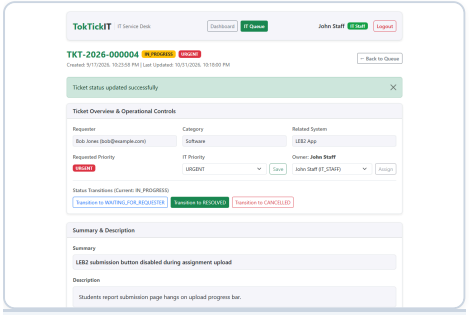



EVD-E2E-02: Full Action Taken Lifecycle (Pending → In Progress → Completed), Assignee Assignment, and Ticket State Reflection.

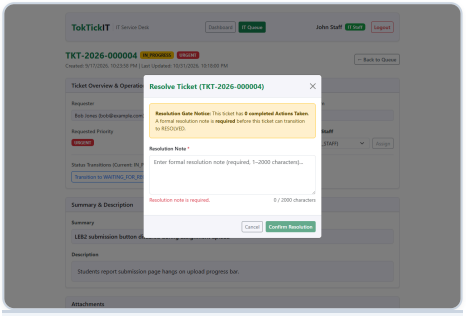
## Answer Part 7: Working Ticket Workflow

5 Points

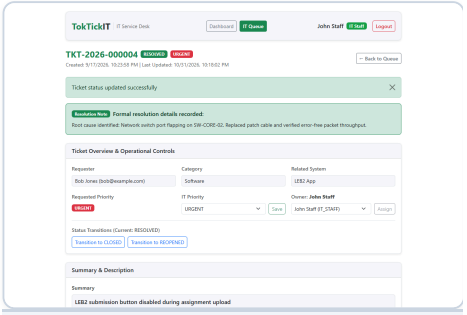
**State Machine & Resolution Gate (tickets.ts / `PATCH /api/tickets/:id/status`):** Enforces 15 allowed transitions, 4 terminal blocks, optimistic concurrency ( `expectedUpdatedAt` ), and the Resolution Gate:



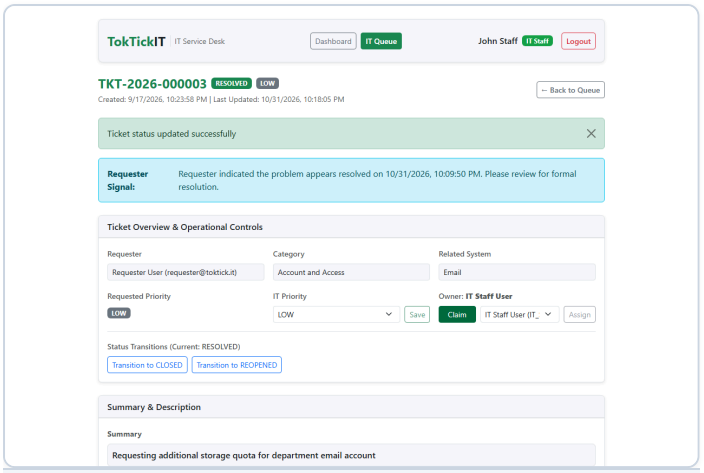
EVD-WORKFLOW-01: Transition from OPEN to IN_PROGRESS.



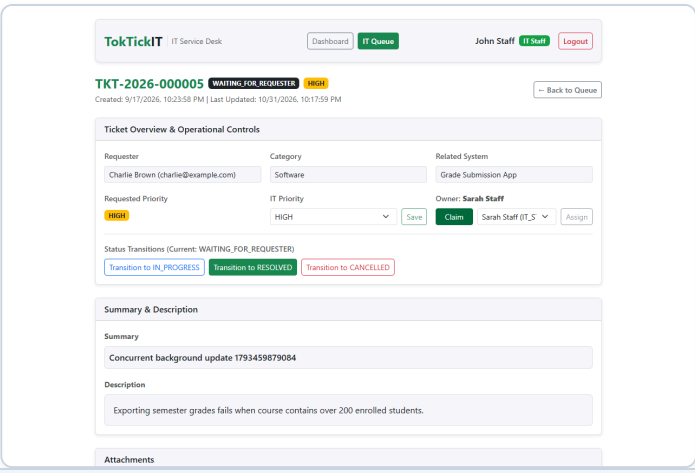
EVD-WORKFLOW-03: Resolution Gate blocks resolution without completed action or resolution note.



EVD-WORKFLOW-04: Successful resolution displaying RESOLVED badge and note.




EVD-E2E-03: Valid Resolution Path #2: Precondition satisfied by completed ActionTaken without requiring note.

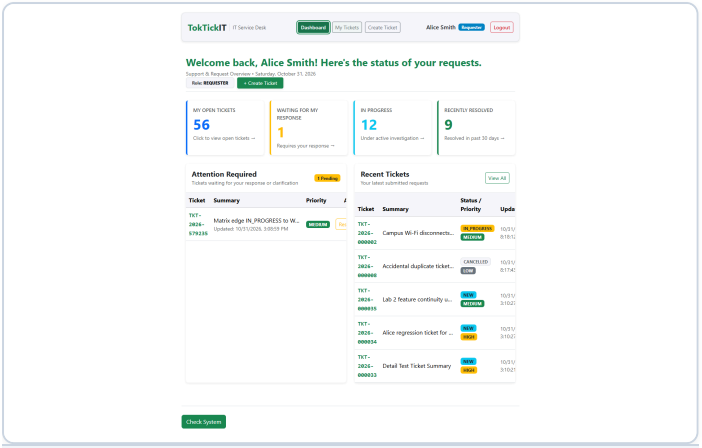



EVD-E2E-05: Concurrency Conflict ( `409 STALE_UPDATE` ) warning banner preserving form inputs with safe refresh.

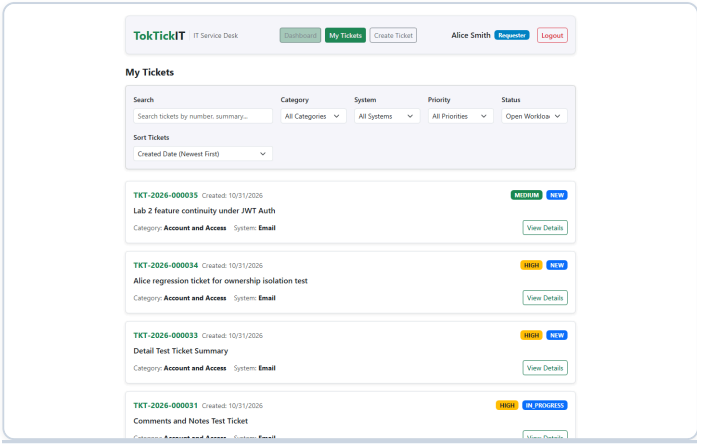
## Answer Part 8: Working Requester Dashboard and Final Regression UI

5 Points

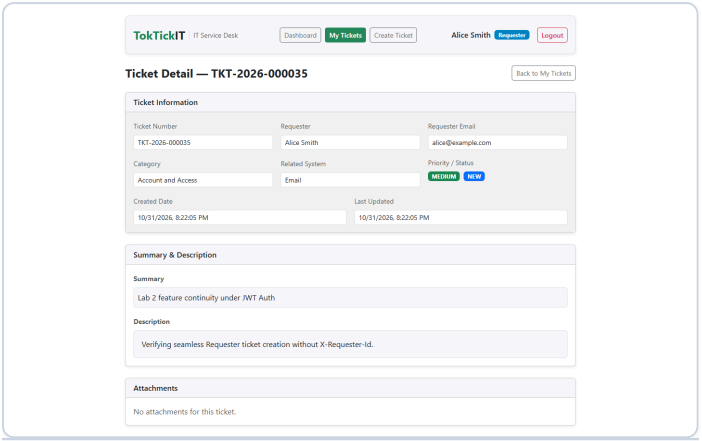
**Requester Dashboard & Full Regression Suite:** Requesters access personalized metrics (My Open Tickets, Waiting for Response, In Progress, Recently Resolved in last 30 days) with strict ownership protection. Full regression guarantees zero regression across Labs 1–3:



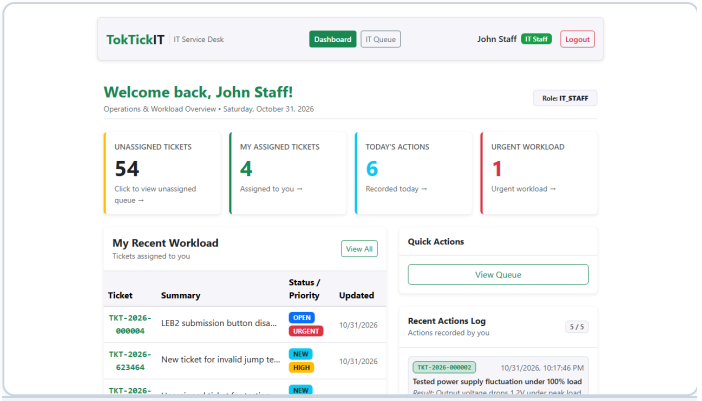

EVD-REQBDASH-01: Requester Dashboard on Desktop (1920×1080) showing personalized metrics and attention worklist.




EVD-REQBDASH-04: Drill-Down Navigation to My Tickets filtering open tickets owned by logged-in Requester.




EVD-REG-02: Consolidated Cross-Role Regression Suite verifying Labs 1–3 features under unified JWT authentication.

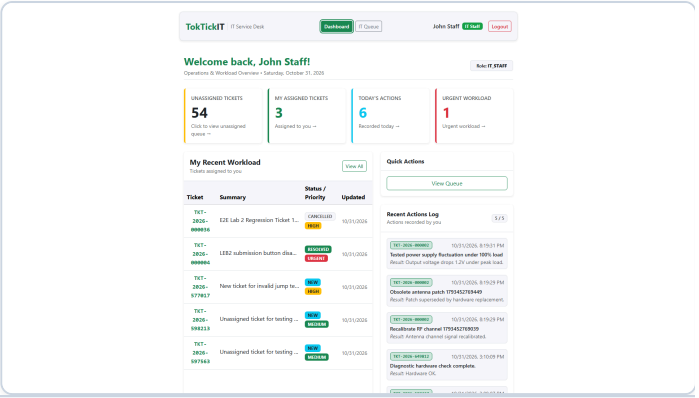



EVD-REG-07: Automated Product Hardening: 0 console errors, 0 broken links, 0 UI placeholders, valid accessible buttons.

## Answer Part 9: Zen Green UI, Responsive, Accessibility, and Final Polish

5 Points

**Visual Design, Multi-Viewport & Accessibility Standards (ui-spec.md):** The application implements the Zen Green design system (#198754 primary, #20c997 accent, #f8fafc neutral background), with verified layout adaptation across Desktop (1920×1080), Tablet (768×1024), and Mobile (375×812) viewports:




EVD-QA-06: Multi-Device Dashboard Responsive Layout Audit (Desktop 4-card, Tablet 2x2, Mobile vertical stack).

## Sprint 4 UI Specification: TokTickIT User Interfaces, Dashboards, and Workspaces

- **System Name**: TokTickIT
- **Sprint / Lab**: Lab 4 (Sprint 4)
- **Document Status**: Final Approved UI Specification (Sprint 4 Release)
- **Repository Branch**: `feature/lab4-visual-qa`

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### 1. Visual Theme & Application Shell

Lab 4 preserves and extends the **Zen Green Design System** introduced in Labs 2 and 3.

#### Design Tokens

- **Primary Brand Color**: Zen Green (`#198754`, `bg-success`, `text-success`)
- **Secondary Accent**: Light Green Tint (`#E8F5E9`, `bg-light`)
- **Status Badges**:

- `NEW`: Info Blue (`bg-info text-dark`)

- `OPEN`: Primary Blue (`bg-primary text-white`)

- `IN_PROGRESS`: Warning Amber (`bg-warning text-dark`)

- `WAITING_FOR_REQUESTER`: Dark Slate (`bg-dark text-white`)

- `RESOLVED`: Success Green (`bg-success text-white`)

- `CLOSED`: Secondary Grey (`bg-secondary text-white`)

- `REOPENED`: Danger Red (`bg-danger text-white`)

- `CANCELLED`: Light Muted (`bg-light text-muted border`)

- **Priority Badges**:

- `URGENT`: Red Badge (`bg-danger text-white`)

- `HIGH`: Amber Badge (`bg-warning text-dark`)

- `MEDIUM`: Green Badge (`bg-success text-white`)

- `LOW`: Grey Badge (`bg-secondary text-white`)

#### Application Header Navigation Shell

- Brand Logo: **TokTickIT** semantic anchor link `/` with subtitle `IT Service Desk` and SPA dashboard transition fallback.
- Role-Appropriate Navigation Links:

- **Requester**: `Dashboard` (`data-testid="nav-requester-dashboard"`), `My Tickets` (`data-testid="nav-my-tickets"`), `Create Ticket` (`data-testid="nav-create-ticket"`).

- **IT Staff**: `Dashboard` (`data-testid="nav-staff-dashboard"`), `IT Queue` (`data-testid="nav-it-queue"`).

- **Administrator**: `Dashboard`, `IT Queue`, `User Management` (`data-testid="nav-user-management"`), and `Ticket Lookup` (permitted administrative utility for direct UUID inspection and operations per Lab 3/4 continuity).

- User Identity Capsule: Shows authenticated User Name, Role Badge (`REQUESTER`, `IT_STAFF`, `ADMINISTRATOR`), and `Logout` action button.
- Focus Indicator: Global `:focus-visible` styling (`outline: 2px solid #198754 !important; outline-offset: 2px;`) ensures accessible keyboard navigation on all interactive elements.
- Touch Targets: Interactive controls on mobile viewports (`<= 768px`) enforce minimum touch dimensions of `44×44px`.
- Development Selector is **strictly removed**.

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### 2. IT Staff Dashboard (`/staff/dashboard`)

#### Layout & Component Structure

1. **Welcome Header**: Displays greeting `Welcome back, [Name]!` with current date.

2. **Operational Metric Cards Grid** (4 Metric Cards):

- **Unassigned Tickets**: Displays count. Subtext: `Click to view unassigned queue`.

- **My Assigned Tickets**: Displays count. Subtext: `Assigned to you`.

- **Today's Actions**: Displays count of actions taken today. Subtext: `Recorded today`.

- **Urgent Workload**: Displays count. Subtext: `Urgent workload`.

3. **Main Content Row**:

- **Left Column (My Recent Workload)**: List of max 5 recent tickets assigned to current staff with ticket number, summary, priority badge, status badge, and drill-down link (`/staff/tickets/:id`).

- **Right Column (Quick Actions & Recent Actions Log)**: Direct action button (`View Queue`) and recent actions taken feed (max 5 actions with ticket link).

#### Drill-Down Navigation Rules

- Clicking **Unassigned Tickets Card** → Navigates to `/staff/queue?view=unassigned_active&owner=unassigned`.
- Clicking **My Assigned Tickets Card** → Navigates to `/staff/queue?view=my_queue_active&owner=my_queue`.
- Clicking **Today's Actions Card** → Navigates to `/staff/queue?view=today_actions`.
- Clicking **Urgent Workload Card** → Navigates to `/staff/queue?view=urgent_workload&itPriority=URGENT`.
- Clicking any ticket row → Navigates directly to `/staff/tickets/:id`.

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### 3. Requester Dashboard (`/requester/dashboard`)

#### Layout & Component Structure

1. **Welcome Header**: Displays greeting `Welcome back, [Name]! Here's the status of your requests.`

2. **Requester Metric Cards Grid** (4 Metric Cards):

- **My Open Tickets**: Displays count of open tickets owned by user.

- **Waiting for My Response**: Displays count of tickets in `WAITING_FOR_REQUESTER`.

- **In Progress**: Displays count of tickets currently being worked on.

- **Recently Resolved**: Displays count of resolved tickets in past 30 days.

3. **Recent Tickets & Attention-Required Worklists**:

- **Attention Required Tickets**: Dedicated list of tickets in `WAITING_FOR_REQUESTER` status with ticket number, summary, priority badge, updated date, and direct action link.

- **Recent Tickets List**: List of max 5 requester's recent tickets with status badge, priority badge, updated date, and direct detail link.

- **Quick Action Button**: `+ Create Ticket` leading to ticket creation form.

#### Dashboard Interaction States

- **Loading State**: Displays skeleton cards or animated loading indicators without triggering page layout shifts.
- **Empty State**: Displays `0` for empty metrics and clear explanatory text for empty lists (e.g. `"No recent tickets"`, `"No tickets require attention"`).
- **Forbidden State**: Displays safe `403 Access Denied` UI feedback if unauthorized access is attempted.
- **Failure State**: Displays generic retryable error alert banner with a `[ Retry ]` button.
- **Not Found State**: Displays safe `404 Ticket not found.` alert banner with prominent direct fallback button `[ Return to My Tickets → ]` allowing the user to gracefully navigate back to the fallback list view if a targeted ticket has been removed.

#### Requester Metric Drill-Down Navigation

- Clicking **My Open Tickets Card** → Navigates to `/tickets/my-tickets?currentStatus=open`.
- Clicking **Waiting for My Response Card** → Navigates to `/tickets/my-tickets?currentStatus=WAITING_FOR_REQUESTER`.
- Clicking **In Progress Card** → Navigates to `/tickets/my-tickets?currentStatus=IN_PROGRESS`.
- Clicking **Recently Resolved Card** → Navigates to `/tickets/my-tickets?currentStatus=RESOLVED&resolvedWithinDays=30`.
- Clicking any ticket row in **Recent Tickets** or **Attention Required** → Navigates directly to `/tickets/:id`.

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### 4. Ticket Detail Actions Taken Workspace

Located inside Ticket Detail (`/staff/tickets/:id` or `/tickets/:id`), positioned below Ticket Information and above Public Comments / Internal Notes.

#### Components & Controls

1. **Workspace Header**: Section title `Actions Taken Work Log` with action counter badge (`X actions recorded`).

2. **Create Action Form** (Collapsible / Modal):

- **Description**: Textarea, 1 to 2000 chars (validated after trimming whitespace), mandatory.

- **Result**: Textarea, 1 to 2000 chars (validated after trimming whitespace), mandatory.

- **Assignee Selector**: Dropdown listing active IT Staff and Administrator accounts (`isActive = true AND role IN ('IT_STAFF', 'ADMINISTRATOR')`). The Assignee Selector is optional and provides an explicit `Unassigned` state (`null`). Inactive/Requester users excluded.

- **Follow-Up Required Switch**: Checkbox (`Follow-up needed`).

## EVD-QA-05: Rendered UI Specification establishing Zen Green color tokens, component typography, and responsive rules.

3. **Actions Taken Feed List**:

- Each action rendered in a structured card with `Action Date/Time`, `Performed By` badge (showing name + role), `Assignee` tag, status badge (`PENDING`, `IN_PROGRESS`, `COMPLETED`, `CANCELLED`), description, result, follow-up note (if present with amber warning border), and attachment notes.

- Actions ordered chronologically by `createdAt ASC`, tie-break `id ASC`.

- Action controls for authorized IT Staff and Administrators: Status transition controls enforcing permitted next statuses per Section 8.2 (`PENDING` → `[ Start Work ]` / `[ Complete Action ]` / `[ Cancel Action ]`; `IN_PROGRESS` → `[ Complete Action ]` / `[ Cancel Action ]`; terminal states `COMPLETED` and `CANCELLED` render no status transition buttons) and `[ Edit ]`. `COMPLETED` and `CANCELLED` `ActionTaken` records disable status transition buttons, but permit content editing via `[ Edit ]` modal using optimistic concurrency (`expectedUpdatedAt`), unless parent Ticket is `CANCELLED`.

- On `CANCELLED` tickets, all Action Taken creation, editing, and status controls are hidden and disabled.

**Action Submission Reliability**

- The client MUST generate one unique `Idempotency-Key` for each intended Action submission.
- The same key MUST be reused when the same submission is retried after a recoverable network failure.
- A new key MUST be generated only for a new user-intended submission.
- The Save Action control SHOULD be disabled/debounced while the initial submission is in flight.
- Backend idempotency remains the authoritative duplicate-submission protection.

**Requester Read-Only View**

- Requesters can view Actions Taken for Tickets they are authorized to access (owned tickets).
- Requesters see Actions Taken in read-only mode.
- Create Action Form, Edit Action, Assignee Selector, Complete Action, Cancel Action, and Status Transition controls are strictly NOT rendered for Requesters.

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### 5. Ticket Workflow & Resolution UI

**Status Controls**

- Render only permitted next-status controls according to Section 8.1 Ticket Status Transition Matrix.
- Disabled or hidden UI controls MUST NOT be treated as the security boundary; server-side validation strictly enforces authorization and transitions.
- `RESOLVED` status transition button requires satisfying the server-side Resolution Gate (>= 1 COMPLETED ActionTaken or non-empty staff `resolutionNote`).
- Resolution Gate failures display a clear inline error message (`"Cannot resolve ticket without at least one completed Action Taken or formal resolution note."`).
- `CANCELLED` renders as a terminal state and exposes no further status controls.
- `409 Conflict` (`STALE_UPDATE`) displays a stale-update alert banner with a `[ Refresh / Reload ]` button to fetch latest ticket state.

**Resolution Note Input Contract**

- When an IT Staff member or Administrator selects the `RESOLVED` status transition, the UI renders a **Resolution Note** text area (1 to 2000 characters).
- If the Ticket already has at least 1 completed `ActionTaken` record, the `resolutionNote` input is optional.
- If the Ticket has 0 completed `ActionTaken` records, the `resolutionNote` input is MANDATORY (validated with trimmed whitespace).
- If submitted without a completed `ActionTaken` and with an empty `resolutionNote`, the UI displays an inline validation error (`"Cannot resolve ticket without at least one completed Action Taken or formal resolution note."`) and blocks request submission.

**Requester Indicators**

- Requester "Problem Appears Resolved" indication (`POST /api/tickets/:id/resolve-indicator`) is advisory and MUST NOT change the Ticket status to `RESOLVED`.
- The indicator records `requesterResolvedIndicatedAt` and allows IT Staff or Administrators to review the Requester's indication before executing a formal Ticket status transition.

**Role Behavior**

- IT Staff and Administrators follow the operation-level authorization matrix and can execute formal status transitions.
- Requesters CANNOT execute formal ticket status transitions; they may post public comments or send the advisory "Problem Appears Resolved" indication.

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### 6. Responsive Breakpoints & Viewport Rules

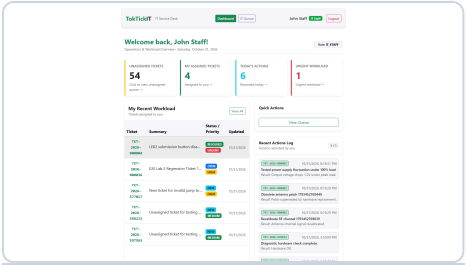
The interface MUST render seamlessly across three standard viewports:

| Viewport Name | Target Dimensions | Responsive Adaptations |
|---|---|---|
| **Desktop** | `1920×1080` (16:9) | Full 4-card metric grid, side-by-side dashboard columns, full data tables with sort headers. |
| **Tablet** | `768×1024` (3:4) | 2×2 metric grid, stacked dashboard columns, horizontally scrollable data tables. |
| **Mobile** | `375×812` (9:19.5) | Single-column metric card stack, touch-friendly action buttons, responsive ticket & action cards. |

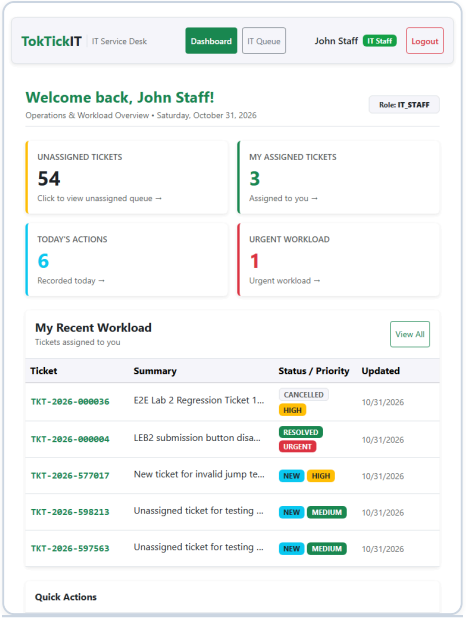
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### 7. Visual & Accessibility QA Checklist

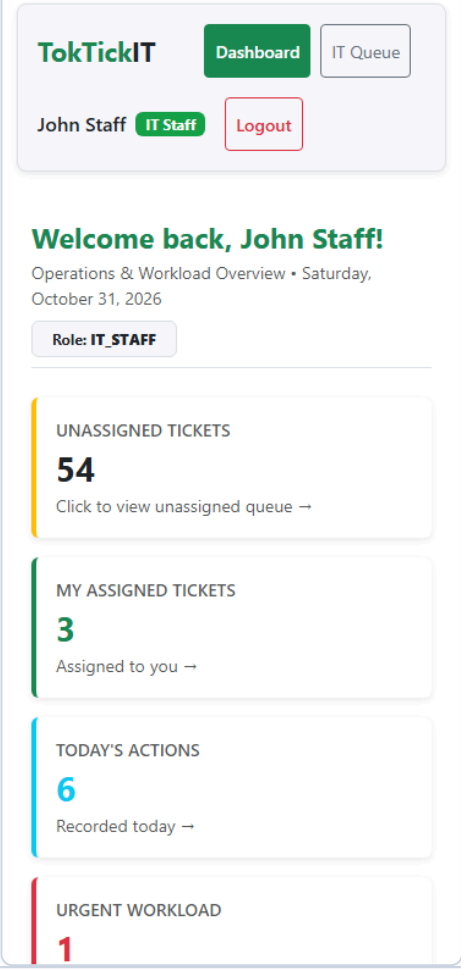
- [x] **No Clipping**: No clipping was observed in the required Lab 4 screens during visual inspection. Automated verification checks page-level horizontal overflow, with targeted bounding-box checks for representative dashboard cards and Actions Taken form controls.
- [x] **No Overlap**: No unintended overlap was observed in the required Lab 4 screens during visual inspection. Automated bounding-box checks cover representative navigation controls, all four Staff Dashboard metric cards, and Actions Taken form fields.
- [x] **No Unintended Horizontal Overflow**: Page body width remains strictly bounded to viewport width (`document.documentElement.scrollWidth <= window.innerWidth + 1`) without relying on body-level overflow clipping (`overflow-x: hidden`). Data tables use isolated responsive scroll wrappers (`overflow-x: auto`).
- [x] **Visible Focus States**: Representative required interactive controls have a visible Zen Green keyboard focus ring (`outline: 2px solid #198754 !important; outline-offset: 2px;`) verified during keyboard traversal.
- [x] **Non-Color Status & Priority Cues**: Statuses, priorities, and roles use distinct readable textual labels alongside color-coded badges, ensuring full accessibility for color-vision-deficient users.
- [x] **Mobile Touch Targets**: Required mobile interactive controls verified in the main navigation and Actions Taken form enforce minimum touch dimensions of `>= 44×44px` on mobile viewports (`<= 768px`).
- [x] **Semantic Labels & ARIA Attributes**: Representative required forms (Login, Create Ticket, Actions Taken) use explicit label-to-control associations (`for`), landmark regions, and valid ARIA roles (`role="region"`, `role="button"`, `aria-current="page"`).
- [x] **Editable vs Read-Only Field Clarity**: Read-only fields (e.g. Requester Actions Taken feed, Ticket ID, timestamps) are visually distinguished from editable controls (Create/Edit Action modals, Resolution Note textarea).
- [x] **Clear Validation & Alert Placement**: Required representative validation states (Login error banner, Actions Taken inline feedback,



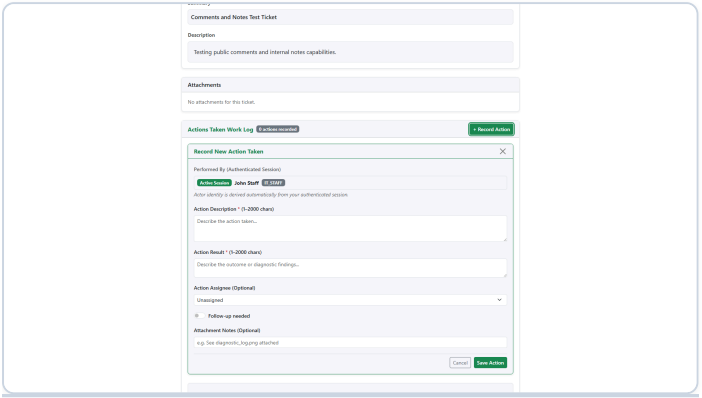
EVD-QA-01: Desktop Viewport (1920×1080)



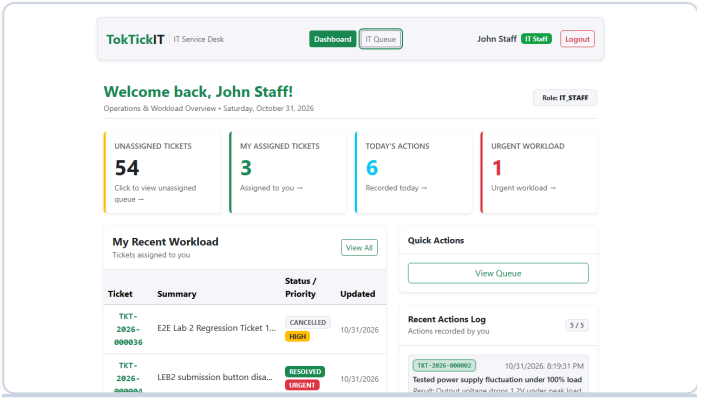

EVD-QA-02: Tablet Viewport (768×1024)




EVD-QA-03: Mobile Viewport (375×812)



EVD-QA-07: Dedicated Actions Taken Multi-Device Responsive Audit & Touch Target Inspection.




EVD-QA-08: Accessibility Focus Rings ( `2px solid #198754` ) with 2px offset and full keyboard tab navigation.

| Accessibility & Quality Check | Specification Invariant | Automated / Visual Verification | Status |
|---|---|---|---|
| **Touch Target Size** | Minimum 44×44px on mobile viewports | `TEST-QA-05` (Mobile inputs & buttons >= 44×44px) | PASS |
| **Focus Indicators** | High-contrast 2px solid #198754 outline | `TEST-QA-06` (Keyboard tab sequence without mouse) | PASS |
| **Color Accessibility** | Non-color textual status/priority cues | `TEST-A11Y-01` (All badges have explicit text) | PASS |
| **Layout & Overflow** | Zero page-level horizontal overflow | `TEST-RESP-01` (0 horizontal scroll at 375px) | PASS |
| **Field Collisions** | No clipping or overlap on narrow screens | `TEST-QA-05` (Bounding-box non-overlap checks) | PASS |
| **Form Preservations** | Safe errors preserve user form inputs | `TEST-FAIL-01` (Inputs intact on 400 error) | PASS |